

Data Analytics Insights Report

Smart City Traffic Intelligence

Executive Summary

Analysis of the Metro Interstate Traffic Volume dataset identified missing, partial and complete duplication of entries. After data cleaning, clear temporal patterns in traffic demand are noted, showing that time of day is a stronger practical indicator of traffic conditions than temperature alone. Traffic volumes were highest during daytime commuting periods, particularly in the late afternoon, while overnight traffic was substantially lower. Weekend traffic patterns differ from weekday. Other factors like temperature and weather conditions have a lesser impact on traffic pattern.

These findings can support smart-city mobility planning by helping traffic-management teams anticipate periods of high demand, allocate resources more effectively and distinguish regular commuting patterns from environmental influences.

Key Findings

1. Data Quality Is Critical to Reliable Traffic Analysis

An important finding from the analysis was that apparent traffic trends can be strongly affected by **data completeness and structure**. The dataset did not provide an equal number of observed hours for every year. For example, 2012 began only in October, while substantial differences in the number of observed hours were also identified in later years. Missing periods therefore need to be considered before interpreting differences in annual traffic totals as genuine changes in traffic demand and their impact to the data set should be addressed to ensure accurate data analysis.

The **timing of missing observations is also important**. Missing observations during low-traffic overnight periods would affect annual traffic statistics differently from missing observations during morning or afternoon peak periods. Consequently, both the amount and the temporal distribution of missing data should be examined before drawing conclusions from traffic trends.

Repeated timestamps provided another important data-quality issue. The importance of distinguishing between **exact duplicate records and repeated or partially duplicated observations** and exploring the reasons for these and their subsequent handling cannot be underestimated. For hourly traffic analysis, using one traffic-volume observation per unique timestamp prevented the same traffic hour from being counted multiple times.

Clues that proved useful for detecting data-quality problems: 1) Unexpectedly large differences between yearly totals or observation counts, extreme or implausible values should also be investigated rather than accepted automatically; for example, temperatures near 0 K produced values close to -273°C after conversion and were identified as erroneous outliers.

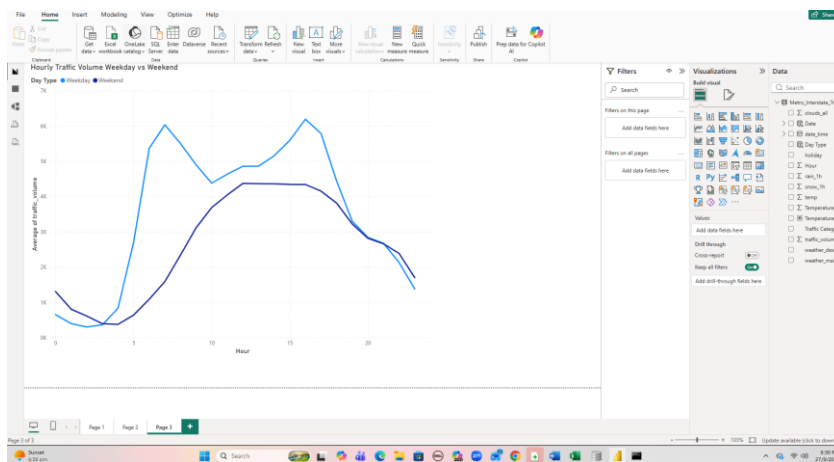
2) Visualisation provided an additional data-validation tool. Plotting traffic over time made large gaps in the dataset easier to recognise. Where long periods contained no observations, a line chart could connect the observations on either side of the gap with an artificial straight line, suggesting missing data.

For a smart-city mobility team, these findings highlight the need for systematic **data-quality checks before operational or strategic decisions are made**. Data completeness, missing observations, duplicates and implausible values should therefore be investigated and appropriately addressed before analysis.

2. Traffic demand is strongly influenced by time of day and patterns differ weekday from weekends

Traffic patterns differed clearly between weekdays and weekends. Weekday traffic shows dual early morning and late afternoon peaks with relatively high flow between the peaks. Weekend traffic showed a different pattern, a gradual increase through the morning and plateauing at 4,300–4,400 vehicles per hour from around midday to late afternoon. A higher weekend traffic in the very early morning is noted compared to weekdays.

This suggests that **day type and hour of day are important determinants of traffic demand**. For smart-city traffic management, weekday demand may require greater capacity and congestion-management measures during the early morning and late afternoon, whereas weekend demand is more broadly distributed across the daytime period.



3. Annual traffic trends require cautious interpretation

Annual traffic totals increased from 2012 to 2013, decreased in 2014, and subsequently increased through 2017. These numbers are greatly affected by the number of entries (2012 data started only in October). The 2014 dataset also contained missing periods and repeated timestamps. Data completeness and coverage need to be considered before using annual totals for long-term transport planning.

4. Temperature alone has only a weak relationship with traffic volume

The Pearson correlation between temperature and traffic volume was approximately **0.137**, indicating only a weak positive linear association, supported by the scatter plot created.

Temperature-bin analysis provided additional context. Average traffic volume is higher at the warmer end of the temperature range, **3,863/3811 vehicles/hour at 30°/35°C**. Again, higher temperature does not imply causation. Other factors such as season, time of day and travel behaviour may come into play.

For mobility planning, temperature should therefore be treated as a supporting variable rather than a primary predictor of traffic demand.

5. Weather conditions are associated with differences in average traffic volume

Average traffic volume varied across weather categories. Among the observed categories, **Clouds recorded the highest average traffic volume at approximately 3,618 vehicles/hour**, while **Squall recorded the lowest at approximately 2,062 vehicles/hour**.

These differences suggest that weather may influence travel patterns. However, uncommon weather conditions may contain substantially fewer observations; so differences in average traffic volume should be interpreted with caution. Further modelling should control for time of day, season and the number of observations within each weather category before attributing changes directly to weather.

6. Holiday temperature differences did not show a clear direct traffic effect

The observed holiday temperature differences therefore do not provide strong evidence that temperature alone materially altered holiday traffic conditions.

Implications for the Smart-City Mobility Team and Conclusion

The analysis indicates that **temporal traffic patterns should be the primary focus of operational traffic management**. Predictable early morning and late afternoon peaks during weekdays provide opportunities for proactive congestion management rather than purely reactive responses.

Weather, temperature information are minor determinants of traffic patterns; need to work together with time in forecasting. A useful next step would be to develop a multivariable traffic-volume forecasting model incorporating temporal, weather and holiday variables. Such a model could support dynamic signal timing, congestion alerts and more efficient allocation of smart-city transport resources.

For smart-city mobility planning, combining reliable time-based patterns with weather and holiday information offers a more robust basis for anticipating congestion and supporting data-driven traffic management.